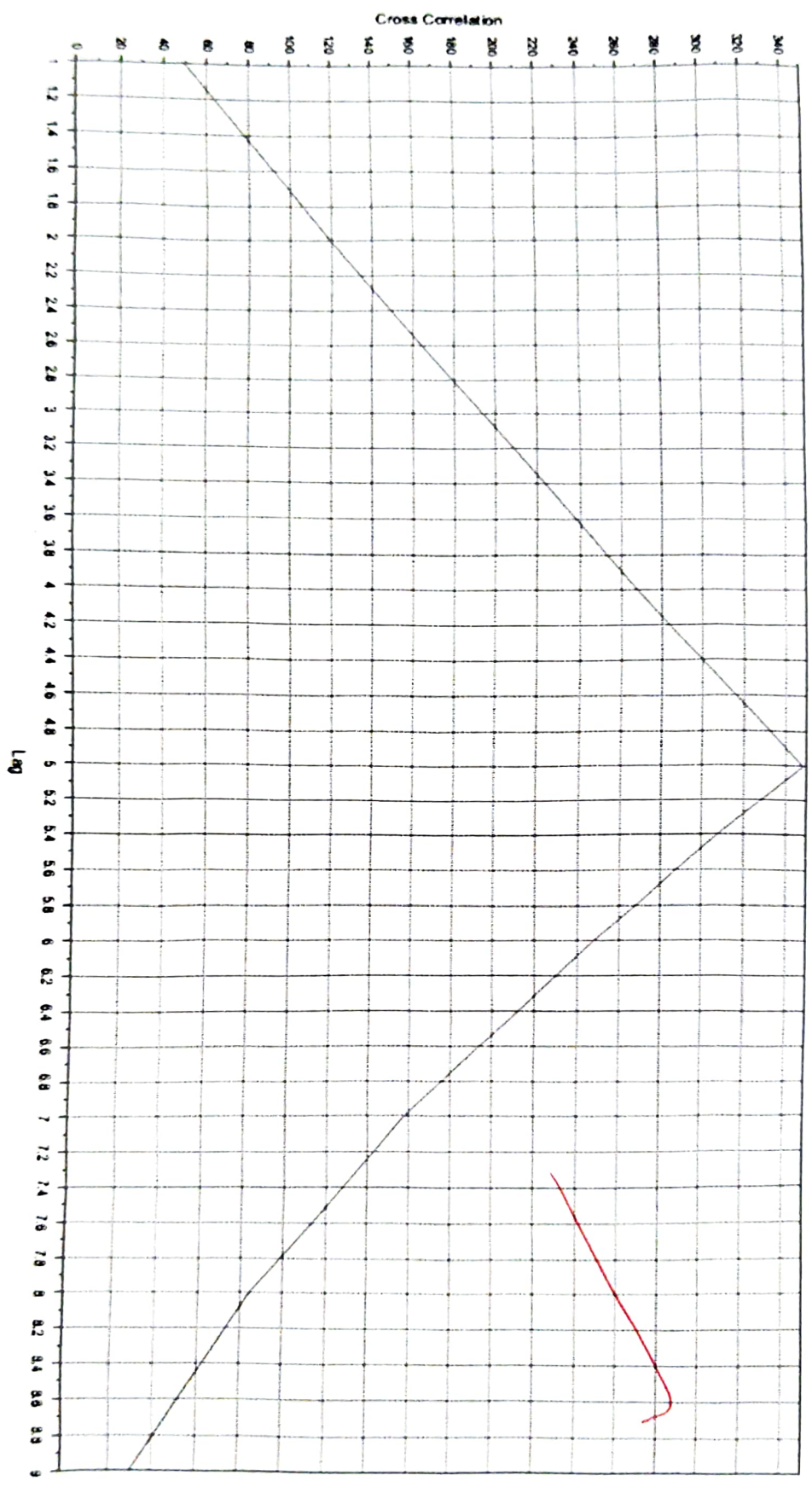


Cross Correlation Waveform



17/08/26

EXPERIMENT 6

CALCULATION OF MEAN AND VARIANCE AND CROSS-CORRELATION

AIM:-

To code and generate the mean and variance for the values in x. and stimulate cross-correlation.

EQUIPMENT REQUIRED:-

Scilab software.

CODE:-

```

clc;
clear;
x = [10 15 20 25 30];
n = 5;
sum_x = sum(x);
mean_x = sum_x / n;
disp("Mean = ");
disp(mean_x);
variance = sum((x - mean_x).^2) / n;
disp("Variance = ");
disp(variance);

```

scilab program & implementation (20)	Waveform and tabulation (20)	calculat ⁿ and result (20)	viva- voce (20)	Result & observation (20)
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```

clc;
clear;
X=[5 8 10 12 15];
Y=[2 4 6 8 10];
N=length(X);
mean_X = sum(X)/N;
variance_X = sum((X-mean_X).^2)/N;
disp(mean_X);
disp(variance_X);
Rxy = xcorr(X,Y);
figure();
plot(Rxy);
xlabel("lag");
ylabel("cross correlation waveform");
fitfit("cross correlation waveform");
xgrid;

```

Scilab program & implementation (20)	Waveform & tabulation (20)	calculation and result (20)	viva- voce (20)	Result & observation (20)
20	20	20	15	20 95

CALCULATION:-

$$\text{Mean} = \frac{\sum x}{n}$$

Add all numbers,

$$10 + 15 + 20 + 25 + 30 \Rightarrow 100.$$

$$\text{Mean}(\bar{x}) = \frac{100}{5} \\ \Rightarrow 20.$$

$$\text{Variance:- } \frac{\sum (x - \bar{x})^2}{N}$$

x	$\bar{x}$	$x - \bar{x}$	$(x - \bar{x})^2$
10	20	-10	100
15	20	-5	25
20	20	0	0
25	20	5	25
30	20	10	100

$$\sum (x - \bar{x})^2 \Rightarrow 250$$

$$\text{So, variance} = \frac{250}{5} \Rightarrow 50.$$

RESULT:-

The mean and variance is generated in scilab. and cross-correlation is generated.

PSV.